# UML Use Case Report: EECS 348 Assignment 4

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## Identification of Use Case

- Use Case Name: Solve Sudoku Puzzles
- Primary Actor: User / System Operator
- Supporting Actor: File System (Puzzle Text Files)

## Brief Description

The Solve Sudoku Puzzles use case explains how the C++ program loads and solves five Sudoku puzzle files, from puzzle1.txt through puzzle5.txt. The program reads each puzzle, checks the rows, columns, and 3x3 blocks, and tries to fill in the missing numbers. Once it finishes, it prints the solved puzzle or displays "No solution found" if it cannot solve it.

## Step-by-Step Flow of Events

1. The program starts and sets up the Sudoku solver.
2. The program goes through the five puzzle files, one at a time.
3. It opens each file and loads the puzzle into a 9x9 grid.
4. It prints the original puzzle before trying to solve it.
5. The program checks the rows, columns, and 3x3 blocks to make sure each number follows the Sudoku rules.
6. It uses recursion and backtracking to try different numbers until it finds a solution or determines that the puzzle cannot be solved.
7. The program prints the completed puzzle or displays "No solution found."
8. It repeats the process until all five puzzle files have been checked.